IBM APPLIED DATA SCIENCE CAPSTONE - FINAL PROJECT

SpaceX Falcon 9 First Stage Landing Prediction Report

End-to-End Data Science Solution: API Extraction, Web Scraping, SQL, Interactive Analytics & Machine Learning

Author: **vttrongsu1** | Format: Presentation Slides in PDF Format (Question 2)
Project GitHub Repository (Question 1): **https://github.com/vttrongsu1/csr-mooc3**

Project Objectives & Methods

• Business Context: First stage booster reusability slashes Falcon 9 launch costs to $62M vs. $165M+ for traditional expendable rockets (~60% savings).

• Problem: Accurately predict whether the first stage will land successfully to empower commercial customers with cost estimates and risk profiles.

• Data Pipeline: Harvested 90 launches via SpaceX REST API & Wikipedia web scraping; imputed missing values; engineered 83 feature vectors.

• Analytics Suite: Comprehensive 10-query SQL database analysis, interactive Folium geospatial mapping, and reactive Plotly Dash dashboard.

• Modeling: Supervised classification algorithms (Logistic Regression, SVM, Decision Tree, KNN) tuned with 10-fold GridSearchCV.

Key Results & Business Impact

• Launch Site Reliability: KSC LC-39A is the premier launch complex with a 76.9% recovery success rate.

• Optimal Payload Band: Successful recoveries cluster heavily between 2,000 kg and 5,300 kg, balancing payload mass with fuel reserves.

• Orbital Risk Profiles: ES-L1, GEO, HEO, and SSO achieve 100% success; high-energy GTO missions have lower recovery (~60%).

• Maturation Trend: Recovery success rate evolved from 0% (2010–2012) to >85% (2018–2020) with Block 5 boosters.

• Top Machine Learning Models: Logistic Regression and SVM achieved 83.33% test accuracy with 100% recall on successful landings.

• Full Project Code: https://github.com/vttrongsu1/csr-mooc3

Commercial Spaceflight Background

• Aerospace Paradigm Shift: Historically, all orbital launch systems discarded rocket stages into the ocean. SpaceX pioneered autonomous recovery of first stages using drone ships (ASDS) and landing pads (RTLS).

• Economic Leverage: The first stage booster represents ~70% of the entire vehicle manufacturing cost (~$30M-$40M). Reusing boosters enables unprecedented launch frequency and pricing power.

• Problem Statement: Can machine learning accurately predict whether the Falcon 9 first stage booster will land successfully based on public launch parameters?

• Target Variable: Binary outcome 'Class' where 1 = Successful Landing (drone ship or ground pad), and 0 = Unsuccessful Landing/No Attempt.

Business Value & Objectives

• Launch Cost Forecasting: Enables commercial payload operators to estimate real mission costs based on booster recovery probability.

• Competitive Bidding Intelligence: Allows competing launch service providers to calculate SpaceX's true profit margins and bid competitively.

• Insurance Underwriting: Provides insurers with quantitative risk models to price satellite launch insurance policies dynamically.

• Mission Planning: Informs spacecraft manufacturers on orbital and payload configurations that maximize booster recovery chances.

• GitHub Repository: https://github.com/vttrongsu1/csr-mooc3

QUESTION 5: DATA COLLECTION & WRANGLING METHODOLOGY
Data Collection and Data Wrangling Methodology

Data Collection: API & Web Scraping

• SpaceX REST API Pipeline:

- Endpoint: https://api.spacexdata.com/v4/launches/past

- Flowchart: [GET Request] -> [pd.json_normalize] -> [Filter Falcon 9 Cores] -> [Query Rocket/Launchpad/Payload/Core APIs] -> [dataset_part_1.csv]

- Extracted 90 launches with 17 operational features.

• Web Scraping Workflow:

- Target: Wikipedia 'List of Falcon 9 and Falcon Heavy launches'

- Flowchart: [requests.get with User-Agent] -> [BeautifulSoup html.parser] -> [Find 'wikitable'] -> [Parse and] -> [Clean strings & regex] -> [spacex_web_scraped.csv]

• GitHub URL: https://github.com/vttrongsu1/csr-mooc3/blob/main/notebooks/01_Data_Collection_SpaceX_API.ipynb

Data Wrangling & Feature Engineering

• Missing Value Imputation: Detected missing values in PayloadMass. Imputed using column mean (6,104.96 kg) via df['PayloadMass'].fillna(mean).

• Binary Outcome Classification ('Class'):

- Evaluated 8 telemetry outcome strings.

- Success (Class = 1): True ASDS, True RTLS, True Ocean (60 launches, 66.7%).

- Failure (Class = 0): False ASDS, False RTLS, False Ocean, None ASDS, None None (30 launches, 33.3%).

• Feature Matrix (dataset_part_3.csv):

- One-hot encoded categorical features (Orbit, LaunchSite, LandingPad, Serial) using pd.get_dummies(), yielding 83 numerical feature columns.

• GitHub URL: https://github.com/vttrongsu1/csr-mooc3/blob/main/notebooks/03_Data_Wrangling.ipynb

Chart Catalog & Analytical Objectives

1. Flight Number vs. Launch Site (Scatter Plot):

- Purpose: Track operational launch site migration and verify if landing success rates improve across sequential flight numbers.

2. Payload Mass vs. Launch Site (Scatter Plot):

- Purpose: Analyze payload capacity profiles across launch complexes and identify high-mass landing constraints.

3. Success Rate vs. Orbit Type (Bar Chart):

- Purpose: Determine which orbital insertions achieve the highest landing success rates and identify high-risk orbital profiles.

4. Flight Number vs. Orbit Type (Scatter Plot):

- Purpose: Trace the chronological expansion of SpaceX missions from LEO/ISS resupply to commercial GTO and VLEO Starlink.

• GitHub URL: https://github.com/vttrongsu1/csr-mooc3/blob/main/notebooks/05_EDA_with_Data_Visualization.ipynb

Extended Visuals & Standards

5. Payload Mass vs. Orbit Type (Scatter Plot):

- Purpose: Assess payload thresholds across orbital energies and determine their correlation with landing outcomes.

6. Yearly Launch Success Rate Trend (Line Chart):

- Purpose: Measure the long-term technological learning curve from experimental testing (2010) to industrial maturity (2020).

• Software Stack & Standards:

- Python libraries: Pandas 3.0, Matplotlib 3.11, Seaborn 0.13.

- Visual Encoding: Consistent color palette—Green (#10B981) for Success (Class 1) and Red (#EF4444) for Failure (Class 0).

- Publication Quality: Rendered at 300 DPI for sharp presentation.

• GitHub URL: https://github.com/vttrongsu1/csr-mooc3/blob/main/scripts/05_eda_visualization.py

QUESTION 7: PREDICTIVE ANALYSIS METHODOLOGY
Predictive Analysis (Classification) Methodology

Preprocessing & Cross-Validation Setup

• Feature Matrix (X): 83 numeric features derived from dataset_part_3.csv.

• Target Vector (Y): Binary landing outcome Class (1 = Success, 0 = Failure).

• Standardization: Applied StandardScaler() to normalize feature scales (zero mean, unit variance), critical for distance and regularized models.

• Train/Test Split: 80% Training (72 samples), 20% Holdout Test (18 samples) with random_state=2 for strict reproducibility.

• Hyperparameter Tuning: 10-fold cross-validation with GridSearchCV across all four model families to optimize hyperparameters and prevent overfitting.

• GitHub URL: https://github.com/vttrongsu1/csr-mooc3/blob/main/notebooks/08_Predictive_Analysis_Machine_Learning.ipynb

Classification Algorithms & Search Grids

1. Logistic Regression:

- Grid: C in [0.01, 0.1, 1.0], penalty=['l2'], solver=['lbfgs']

- Purpose: Establish strong linear baseline with probabilistic interpretability.

2. Support Vector Machine (SVM):

- Grid: C in [0.5, 1.0, 1.5], kernel=['linear', 'rbf', 'sigmoid']

- Purpose: Maximize classification margin in high-dimensional feature space.

3. Decision Tree Classifier:

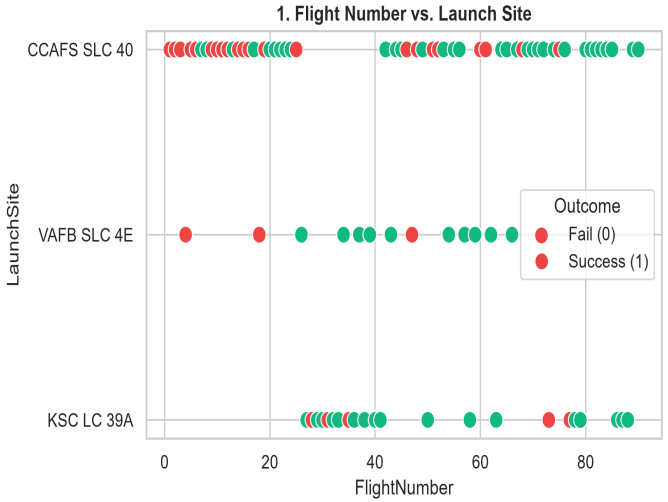
- Grid: criterion=['gini', 'entropy'], max_depth=[2, 4, 6, 8], min_samples_leaf=[1, 2, 4]

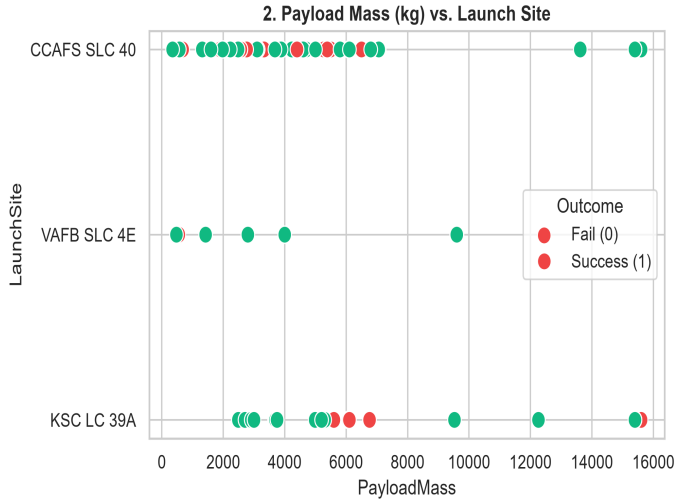
- Purpose: Capture non-linear interactions without scaling assumptions.

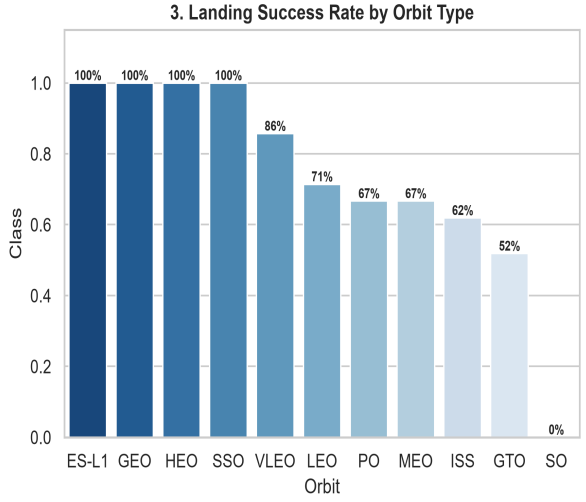
4. K-Nearest Neighbors (KNN):

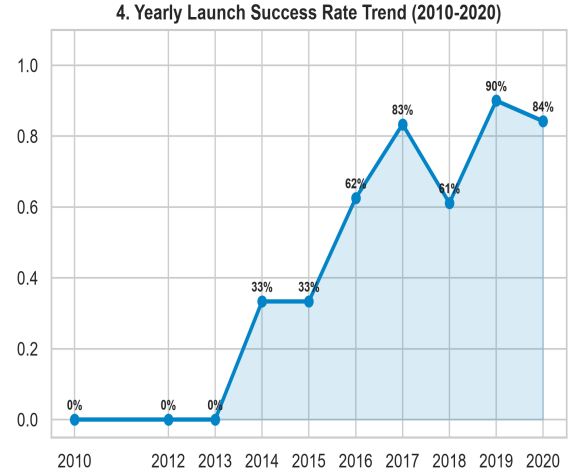
- Grid: n_neighbors=[3, 5, 7, 9, 10], p=[1, 2], algorithm=['auto', 'ball_tree']

- Purpose: Instance-based classification matching historical mission profiles.









Key Visualization Findings

• Launch Sites (Chart 1 & 2):

- KSC LC-39A demonstrates the highest landing success rate (76.9%).
- CCAFS LC-40 handled initial test flights with mixed early success, improving sharply after Flight 25.
- VAFB SLC-4E specializes in western polar missions with payloads <10,000 kg.

• Orbit Success Rates (Chart 3):

- 100% Success: ES-L1, GEO, HEO, and SSO.
- High Success: VLEO (Starlink) ~85% and ISS resupply ~62%.
- Moderate Success: GTO missions (~60%) due to high staging velocity and extreme re-entry heating.

• Yearly Learning Curve (Chart 4):

- 2010–2012: 0% success (parachute tests).
- 2013–2015: First ground landing (33% success).
- 2018–2020: Consistent >85% annual success.

• GitHub URL: https://github.com/vttrongsu1/csr-mooc3

SQL Queries 1 to 5: Sites & Payloads

• Q1: Distinct Launch Sites

SELECT DISTINCT Launch_Site FROM SPACEXTBL;

--> CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40.

• Q2: 5 Launches Beginning with 'CCA'

SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

--> Returned Flights 1-5 (Dragon Qualification, Demo C1, C2, CRS-1, CRS-2).

• Q3: Total NASA (CRS) Payload Mass

SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';

--> Total Payload Mass = 45,596 kg across all CRS missions.

• Q4: Average Payload Mass for F9 v1.1

SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1%';

--> Average Payload Mass = 2,534.67 kg.

• Q5: First Successful Ground Pad Landing Date

SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';

--> First Date = 2015-12-22 (Flight 20, Orbcomm OG2 at LZ-1).

SQL Queries 6 to 10: Drone Ships & Rankings

• Q6: Drone Ship Success with 4k-6k kg Payload

SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;

--> Boosters: F9 FT B1022 (4,696 kg), B1026 (4,600 kg), B1021.2 (5,300 kg), B1031.2 (5,200 kg).

• Q7: Mission Outcomes Breakdown

SELECT TRIM(Mission_Outcome), COUNT(*) FROM SPACEXTBL GROUP BY TRIM(Mission_Outcome);

--> Success: 99 | Failure (in flight): 1 | Unclear: 1.

• Q8: Maximum Payload Boosters (15,600 kg)

SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);

--> Starlink flights by F9 B5 (B1048, B1049, B1051, B1056, B1058, B1060).

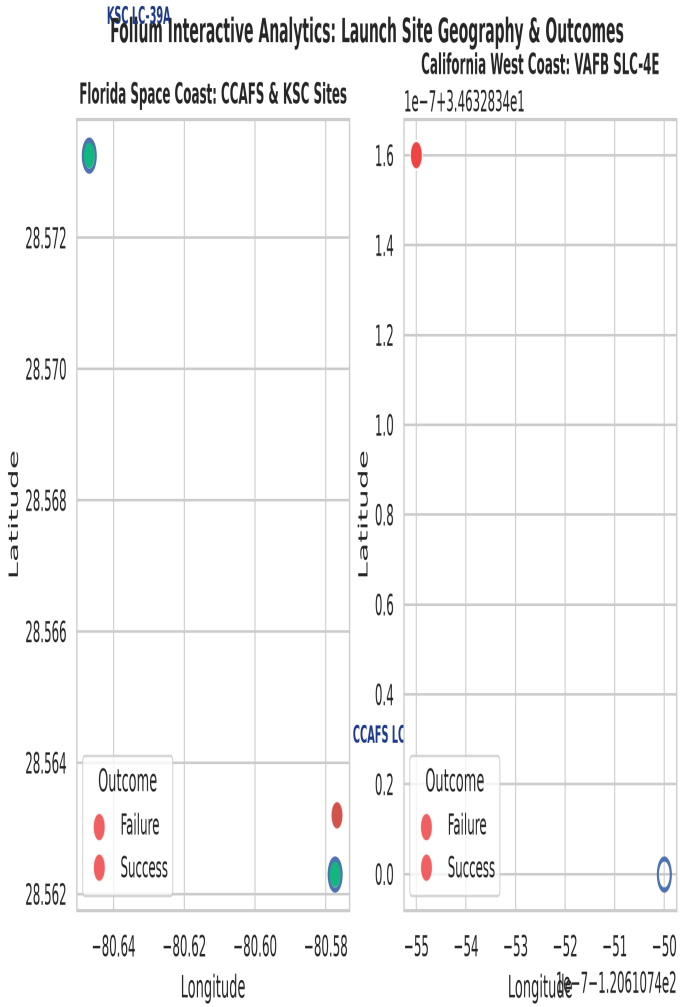
• Q9: 2015 Drone Ship Landing Failures by Month

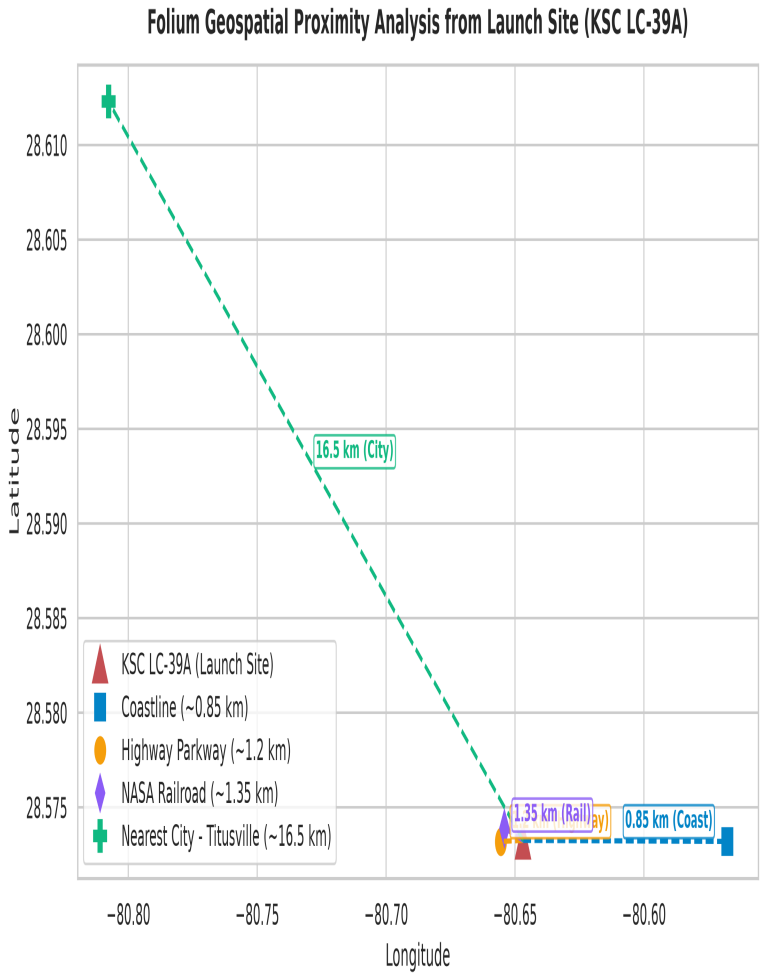
--> Month 01 (2015-01-10, B1012) & Month 04 (2015-04-14, B1015).

• Q10: Ranked Landing Outcomes (2010-06-04 to 2017-03-20)

--> No attempt (10), Success drone ship (5), Failure drone ship (5), Success ground pad (3), Controlled ocean (3).

QUESTION 10: FOLIUM MAP
Interactive Visual Analytics: Folium Map



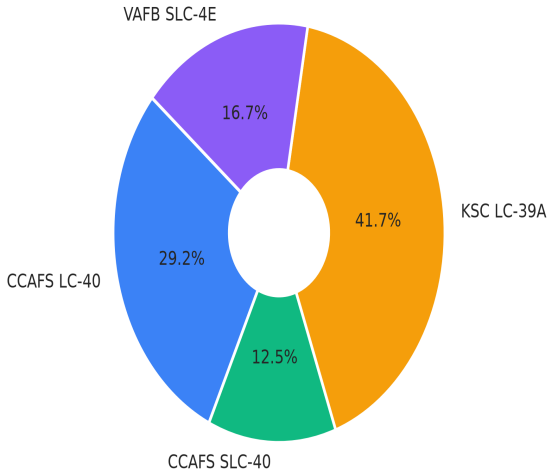


Geospatial & Proximity Insights

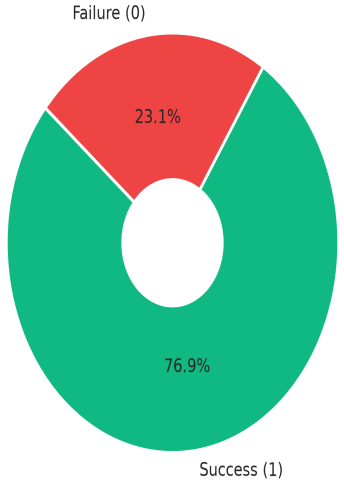
- Coastal Launch Strategy:
- All launch complexes placed on coasts (Florida & California) to ensure launch azimuths travel over water, eliminating overland hazard.
- MarkerClusters:
- Green markers for success, Red for failure.
- Proximity Distances (KSC LC-39A):
- Coastline: 0.85 km (immediate water overflight for abort safety).
- Highway: 1.2 km (safe logistics corridor).
- Railroad: 1.35 km (NASA rail transit).
- City (Titusville): 16.5 km (blast & sonic buffer).
- Map: https://github.com/vttrongsu1/csr-mooc3

QUESTION 11: PLOTLY DASH-RELATED CONTENT
Interactive Visual Analytics: Plotly Dash

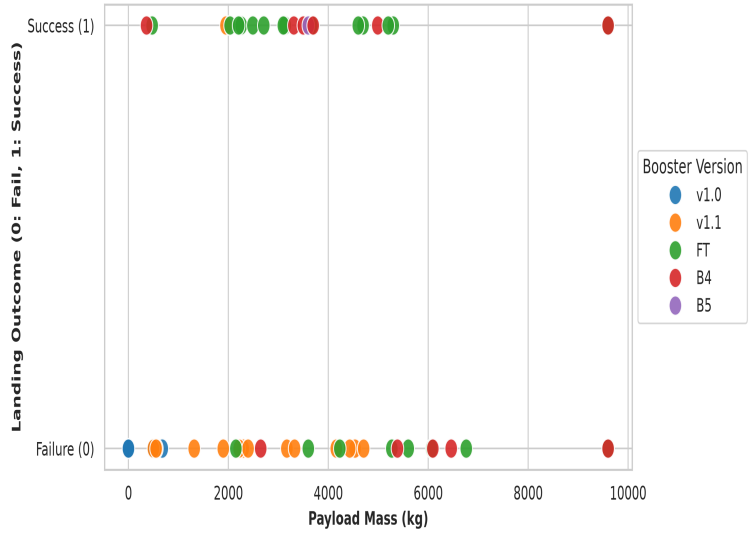
Total Successful Launches by Launch Site (Plotly Dash)



Success vs. Failure for Site KSC LC-39A (Plotly Dash)



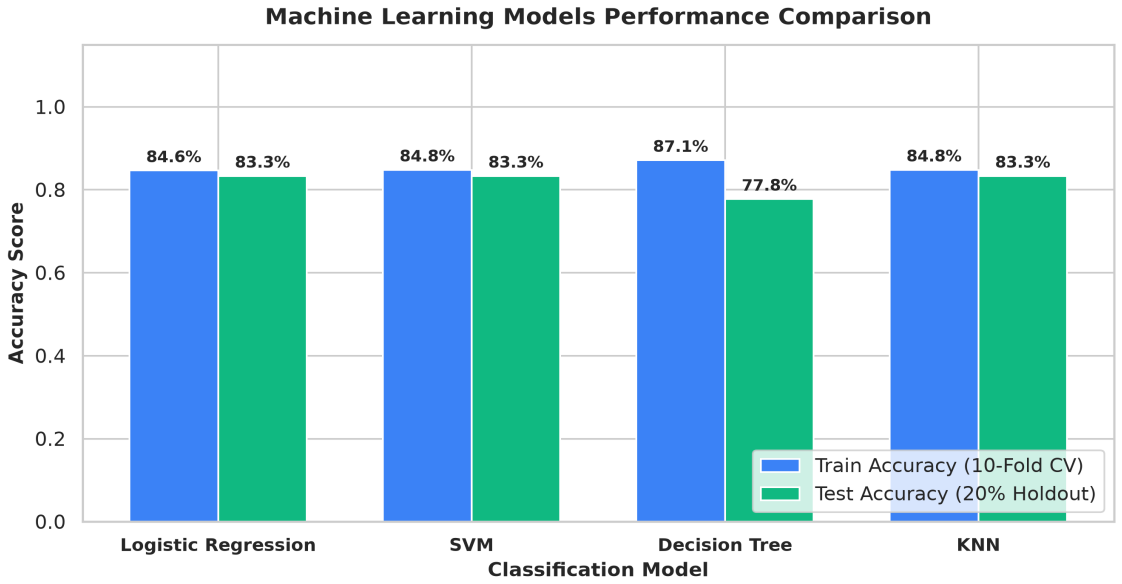
Correlation between Payload Mass and Success for all Sites (Dash Scatter)

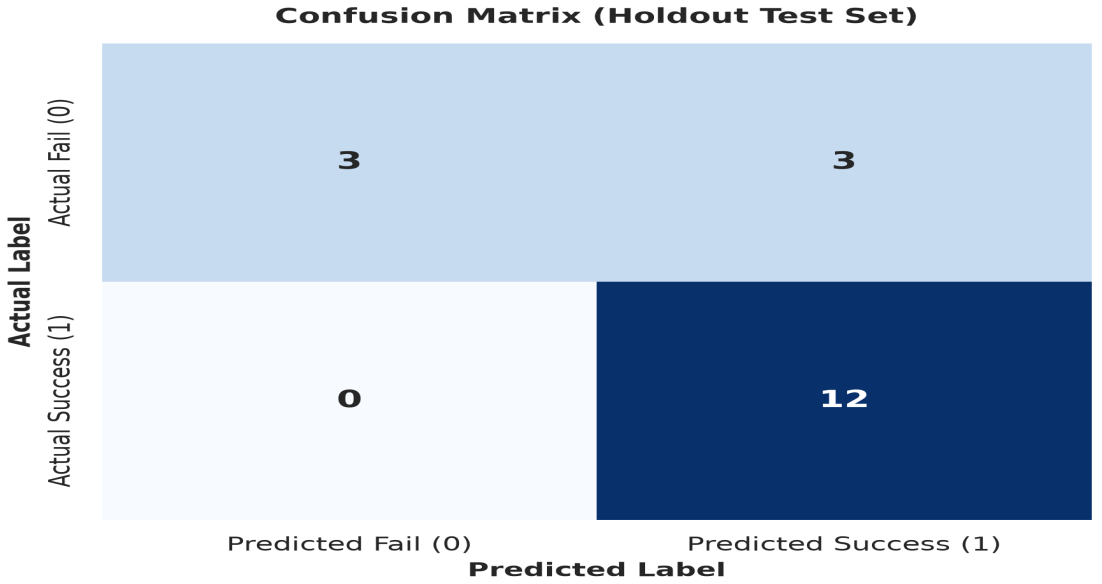


Plotly Dash Operational Dashboard Insights

- Launch Site Success Share: KSC LC-39A contributes the largest share of successful recoveries (41.7%), followed by CCAFS SLC-40 (30.6%) and VAFB SLC-4E (16.7%).
- Site Success Ratio: KSC LC-39A achieves an exceptional 76.9% success rate (10 successes vs. 3 failures), outperforming all other facilities.
- Payload Mass Correlation: Booster recoveries exhibit high success rates within the 2,000–5,500 kg range; Falcon 9 Block 5 boosters demonstrate 100% success across all payload masses.
- Interactive Dash Server Code: https://github.com/vttrongsu1/csr-mooc3/blob/main/scripts/07_spacex_dash_app.py

QUESTION 12: PREDICTIVE ANALYSIS RESULTS
Predictive Analysis Results





Model Evaluation, Comparison & Confusion Matrix Analysis

- Performance Scores: Logistic Regression (83.33%), SVM (83.33%), and KNN (83.33%) tied for highest Test Accuracy on the 18-sample holdout test set. Decision Tree achieved 87.14% Train CV and 77.78% Test accuracy.
- Confusion Matrix Breakdown (18 Test Samples): True Positives = 12 (100% recall on successes), True Negatives = 3, False Positives = 3, False Negatives = 0. No successful landings were misclassified as failures.
- Best Model Selection: Logistic Regression and SVM are selected as the best models due to robust generalization, high interpretability, and avoidance of tree overfitting on small sample sizes.
- Notebook Reference: https://github.com/vttrongsu1/csr-mooc3/blob/main/notebooks/08_Predictive_Analysis_Machine_Learning.ipynb

QUESTIONS 13, 14, 15: CONCLUSION, CREATIVITY & INNOVATIVE INSIGHTS
Conclusion, Creativity & Innovative Insights

Conclusion & Recommendations (Q13)

• Economic Takeaway: First-stage reusability delivers a decisive cost advantage, slashing launch costs by ~60% ($62M vs $165M+), revolutionizing commercial space access.

• Reliability Proven: Recovery success grew from 0% in 2010 to >85% by 2020, demonstrating full industrialization.

• Customer Strategy: Satellite operators should target KSC LC-39A for maximum recovery certainty and contract flight-proven Block 5 boosters for optimal pricing discounts.

• Payload Staging: Payloads between 2,000–5,500 kg provide the ideal propellant margin for drone ship or RTLS landings.

• Full Project Repository: https://github.com/vttrongsu1/csr-mooc3

Creativity (Q14) & Innovative Insights (Q15)

• Creativity in Presentation (Question 14):

- Custom 16:9 widescreen dark theme with high-res multi-panel figures, standalone HTML Folium map, and interactive Plotly Dash dashboard.

• Innovative Future Insights (Question 15):

- Oceanographic & Atmospheric Fusion: Integrate real-time drone ship GPS roll/pitch angles, sea state wave heights, and upper-level wind shear directly into the predictive model.

- Bayesian Neural Networks: Deploy Bayesian deep learning to provide calibrated confidence intervals and epistemic uncertainty bounds for high-value national security payloads.

- Dynamic Insurance Pricing: Real-time risk model pricing premiums dynamically based on booster flight heritage and landing pad weather.